Uday Raj

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Education

Indian Institute Of Information Technology, Design and Manufacturing **Kurnool**
Bachelor of Technology in Artificial Intelligence and Data Science - ***Grade : 8.38*** *2024 - 2028*

Delhi Public School Sushant Lok **Gurugram**
High School - (Physics, Chemistry, Mathematics) - ***Grade : 95%*** *2022 - 2024*

Experience

SVNIT Surat **Nov 2025 - Apr 2026**
Research Intern - Department of Artificial Intelligence *Surat, India*

- Designed **TransEvade** under Dr. Praveen Kumar Chandaliya, a printability-constrained attention-hijacking attack on transformer-based Person Re-ID systems; authored the paper, now under review at an **A-ranked conference**.
- Analyzed a SegFormer/MiT Re-ID pipeline on Market-1501, DukeMTMC-reID and MSMT17 (MiT-B0/B1, 256/512-D descriptors), optimizing 500-1000 step adversarial patterns with masked distract-suppress attention losses.
- Quantified 90-identity spatial ablations where torso patches cut adversarial Rank-1 to **1.67%-7.04%** on Duke/MSMT, exposing attention priors.

Projects

CourtSense AI - Tennis Video to 3D Replay Engine | *YOLOv8, OpenCV, Kalman Filter, FastAPI, Three.js* [**GitHub**]

- Built a 6-stage async CV pipeline (ingest, court calibration, player and ball tracking, analytics) that turns match video into 30 FPS 3D replays, using YOLOv8-Pose (17 keypoints) and HSV + Hough court-line homography.
- Stabilized tracks with a 3D Kalman filter, One-Euro smoothing and spline gap-filling; when ball detections cover under 10% of frames, a physics-based rally synthesizer (gravity, 0.75 bounce restitution) fills the trajectory.

Anatomy of Adversarial Attacks on Person Re-ID | *PyTorch, ResNet-50, DenseNet-121, ViT-S* [**GitHub**]

- Implemented FGSM, PGD and MI-FGSM from scratch on Market-1501; one 64x48 universal torso patch cut ResNet-50 Rank-1 from **86.97% to 2.67%** (random-noise control: 73.19%), transferring to DenseNet-121 at 13.63%.
- Adversarial training raised white-box PGD Rank-1 from 0% to 29.19% at a 12.45-point clean-accuracy cost, with every reported number traced to logged runs and checkpoint hashes.

PaliGemma Vision-Language Model From Scratch | *PyTorch, SigLIP, Gemma* [**GitHub**]

- Rebuilt PaliGemma in pure PyTorch: a 12-layer SigLIP ViT (224x224 into 196 tokens), a 1.6M-parameter 768→2048 projector, and a Gemma decoder with RMSNorm, RoPE, grouped-query attention and KV-cached decoding.

DistroSync - Distributed Task Queue & Message Broker | *Python, asyncio, TCP, SQLite* [**GitHub**]

- Built a broker-free task queue over raw asyncio TCP with SQLite WAL crash recovery and token-bucket load shedding; moved 10,000 tasks from 50 producers to 4 workers at ∼**6,000 tasks/sec, 4.9 ms P99**.

Technical Skills

Languages: Python, C, C++, SQL, TypeScript, JavaScript
Computer Vision: YOLOv8, Pose Estimation, OpenCV, Object Tracking, Kalman Filtering, Homography, ViT, SegFormer
Adversarial ML & GenAI: PGD, MI-FGSM, EOT Patches, Adversarial Training, VLMs, RoPE, GQA, KV-Cache
ML Stack: PyTorch, TensorFlow, scikit-learn, NumPy, Pandas, CUDA, TensorRT, Jupyter, Kaggle
Backend & Tools: FastAPI, asyncio, REST APIs, Docker, Linux, Git, GitHub Actions, Next.js, Vercel

Achievements

- **Global top 20** in the [**Shell.ai**] Hackathon (1000+ submissions); **176th of 7000+** in AWS x Zelestra ML Ascend.
- Qualified for **Amazon ML Summer School** – top 3,000 of 1,30,000+ registrants nationwide.
- Solved **200+** competitive programming problems across platforms; [**CodeChef**] 3-star (1631), [**Codeforces**] Pupil (1314).

Positions of Responsibility

Google Developer Groups
Machine Learning Coordinator *Aug 2025 - Present*

- Ran a 12-week ML bootcamp for 200+ students (SGD, CNNs, Kaggle workflows), resulting in 15+ student projects.

DataWorks Club
Computer Vision Coordinator *Aug 2025 - Present*

- Mentored 100+ members via OpenCV/PyTorch workshops on real-time YOLOv8 + TensorRT pipelines (30+ FPS).